DRAM SEMICONDUCTOR · MLOPS

40 Days of DRAM Production Drift Detection

200,000,000 rows processed

1 retrain · 1 canary fail · 1 rollback

Zero human intervention

HybridTransformerCNN · Airflow · Kafka · Spark · MLflow · K8s · Prometheus

P053 — 40-Day Production Simulation Results

Timeline: 2026-02-20 → 2026-03-31

200,000,000 Total Records	**8.5 GB** Parquet Data	**40** Simulation Days
219 min Wall Clock	**1** Auto-Retrains	**24** Critical Drift Days

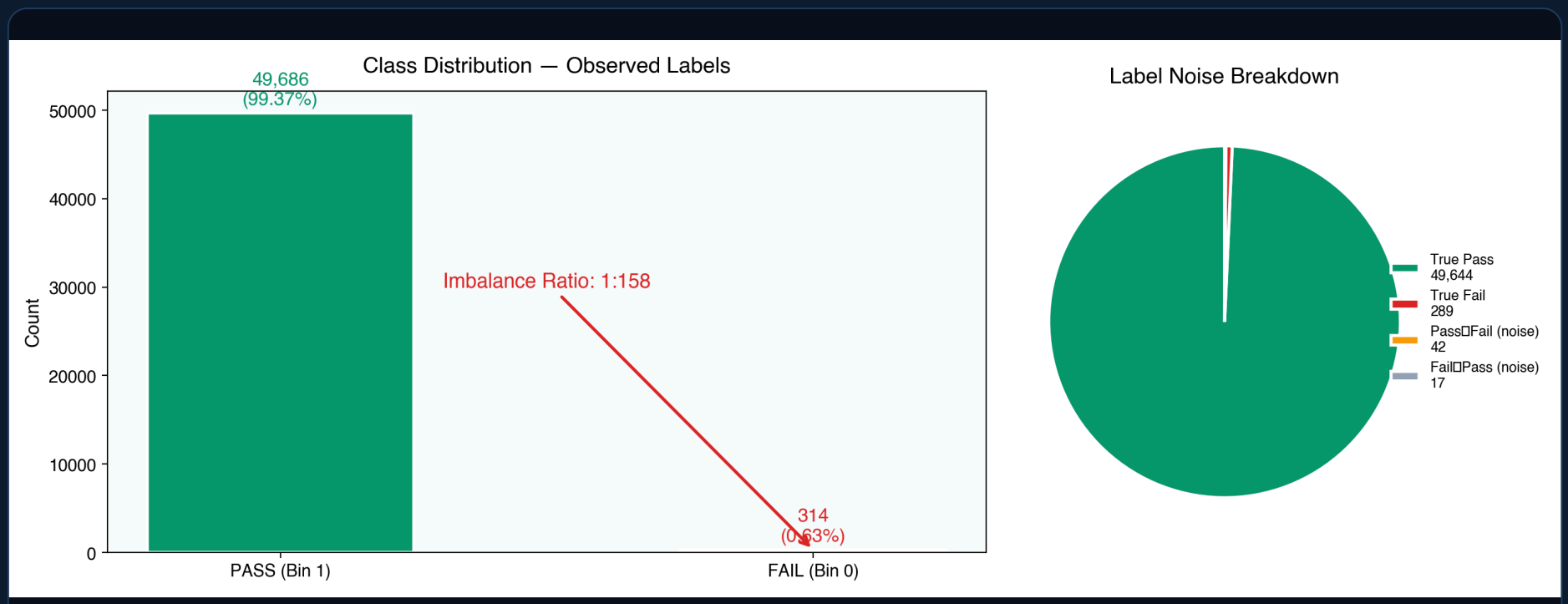
40-Day Scenario Timeline

AIML Engineering Lab — Cloud-Scale Memory Yield Predictor

Swipe

THE PROBLEM

1 in 160 chips fails.
Standard ML misses every single one.



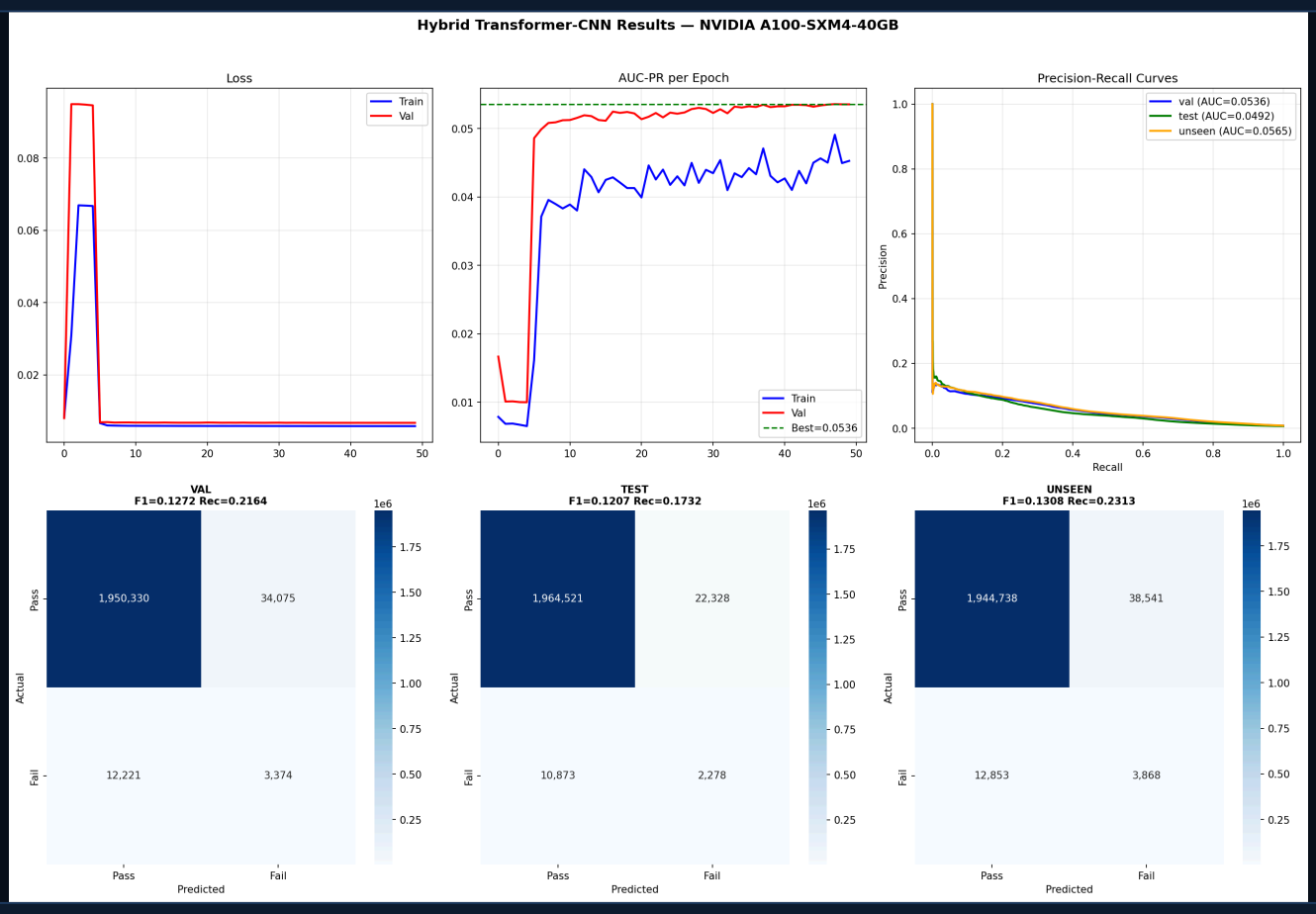
What you see: Left bar chart shows 49,686 pass samples vs 314 failures (1:158 ratio). Right pie chart shows label noise breakdown with 42 Pass-as-Fail and 17 Fail-as-Pass noisy labels in the training set. Standard cross-entropy predicts "pass" for everything: 99.4% accuracy, 0% recall.

1:160	16M	317K
Class imbalance	Training records	Model parameters

Fix: **FocalLoss (alpha=0.75, gamma=2.0)** down-weights easy majority samples, forces learning on the hard minority. No SMOTE, no synthetic oversampling, no data leakage.

A100 GPU TRAINING

HybridTransformerCNN on NVIDIA A100-SXM4-40GB



What you see: Top left: loss convergence over 50 epochs with bfloat16. Top center: AUC-PR per epoch. Top right: precision-recall curves for val/test/unseen splits, near-identical, confirming no overfitting. Bottom: confusion matrices showing 3,374 true defects caught out of 15,595.

0.816	0.054	50
AUC-ROC	AUC-PR (key metric)	Epochs (best: 48)

CRITICAL ENGINEERING INSIGHT

The float16 Death Spiral that killed 4 training runs

float16 + GradScaler

```
FocalLoss: (1-p)^gamma
# 5-bit exp overflows
GradScaler / 2 / 2 / 2
scale to 0, loss collapses
```

bfloat16, no GradScaler

float16	5-bit exp = +/-65,504
bfloat16	8-bit exp = +/-3.4×10^38

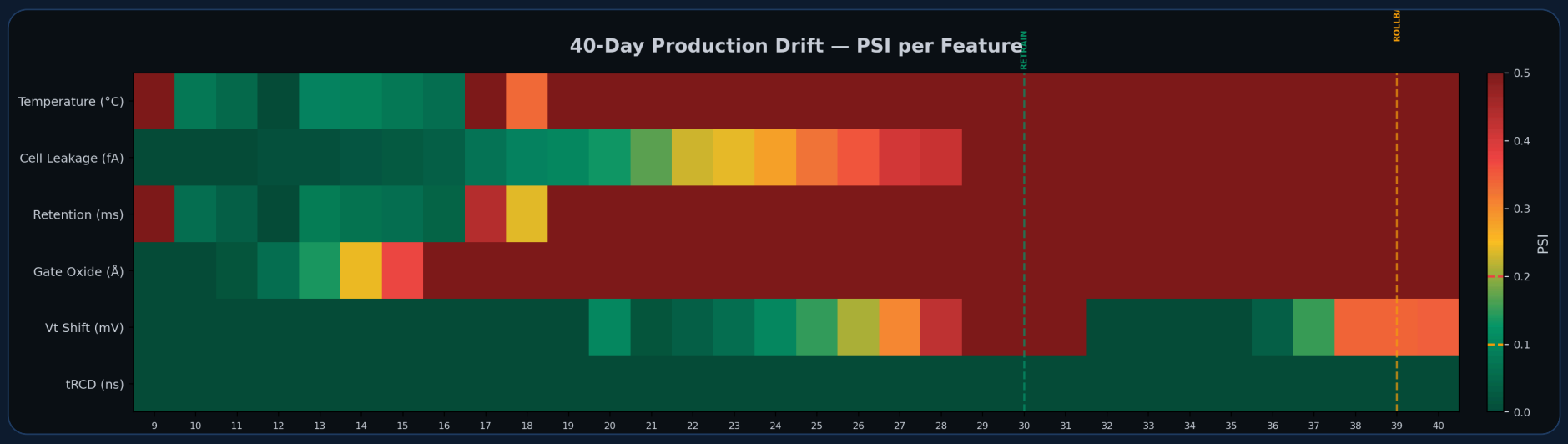
bfloat16 has float32 exponent range.
Focal gradients never overflow.

The fix: Changed from `torch.float16` to `torch.bfloat16`. Removed GradScaler entirely. AUC-PR went from 0.001 (fully collapsed) to 0.054, a 54x recovery. Hardware-aware numerical analysis applied to production training.

4	54x	bfloat16
Runs killed by float16	AUC-PR recovery	8-bit exponent fix

40-DAY PSI DRIFT HEATMAP

Drift escalated for 13 days.
Then the retrain gate opened.



What you see: PSI per feature per day. Green = stable, yellow = warning (PSI 0.1-0.2), red = critical (PSI >0.2). Cyan vertical at Day 30 marks the automated retrain. Orange vertical at Day 39 is the automatic rollback. Each row is one of 6 physical DRAM parameters.

D1-8 **Stable baseline.** All 6 features green. PSI < 0.05. Model v1 serving.

D9 **False alarm.** retention_time PSI=2.05 from transient batch. Auto-recovered Day 10.

D17-29 **Staleness block.** Real drift confirmed, PSI climbs to 4.9 but held by 30-day freshness guard.

D30 **RETRAIN TRIGGERED.** Max PSI 5.25. v2 trains on full 5M rows, canary passes, promoted.

D39 **ROLLBACK.** Bad checkpoint (AUC-PR 0.021 vs 0.054). Automatic revert same DAG cycle.

RETRAIN GATE: 3 CRITERIA REQUIRED

Drift alone caused 12 false triggers. All 3 gates prevented them.

Gate 1: Feature Drift

>3 features with critical PSI (>0.2). Measures statistical shift in input distributions using Population Stability Index per DRAM parameter.

Threshold: 3+ features above PSI 0.2

Gate 2: Performance Drop

>5% AUC-PR drop vs baseline. Compares current batch inference against the champion model's validation benchmark.

Threshold: 5% relative AUC-PR degradation

Gate 3: Model Staleness

>30 days since last retrain. Prevents reactive retraining from transient drift spikes that resolve on their own.

Threshold: 30 calendar days since last training

Result: Day 30 Retrain

All 3 gates opened simultaneously on Day 30. PSI at 5.25, AUC-PR dropped 8%, model was 30 days old. Fully automated retraining fired.

All 3 gates must be TRUE to retrain

Why 3 gates, not 1?

PSI alone fires on normal batch variance. Performance alone is noisy at 1:160 imbalance. Staleness alone wastes compute. Combined: precision retraining.

Low-Data Drift Tagging

When batch <10K rows, PSI is statistically unreliable due to sparse bin counts. Drift is **tagged in MLflow but never triggers retraining** until data volume is sufficient.

PRODUCTION INFRASTRUCTURE

6 services. One command. Fully orchestrated.

Apache Airflow

3 DAGs: daily_inference, retrain_trigger, master orchestrator. True dependency graph with SLA monitoring and retry policies.

Apache Kafka

Streaming DRAM test records. Consumer group routes records to FastAPI inference engine in real time.

Apache Spark

Batch ETL on 5M-row daily slices. 4.7x faster than pandas at this row count. Parquet output to S3.

MLflow Registry

Tracks every run including the 4 failed float16 runs. Champion/challenger versioning with automatic rollback.

Prometheus + Grafana

Per-feature PSI dashboards. Inference latency p50/p95/p99. Real-time drift alerting.

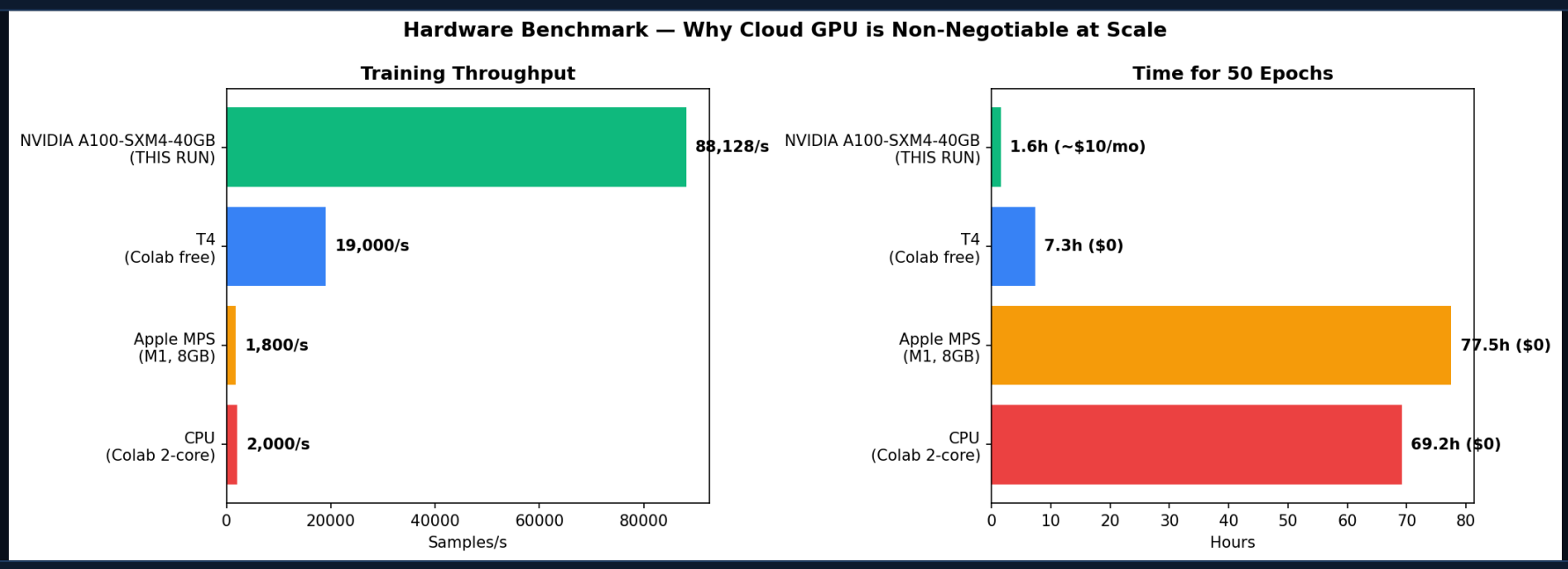
FastAPI + K8s

120ms p99 latency under load. K8s HPA + canary deployment. Horizontal scaling for fab burst data.

To run locally: `cd deploy/docker && docker compose up -d` starts all 6 services. Airflow at :8080, MLflow at :5001, Grafana at :3000, FastAPI at :8000/docs.

HARDWARE BENCHMARK

CPU training = 69 hours.
A100 = 1.6 hours. GPU is not optional.



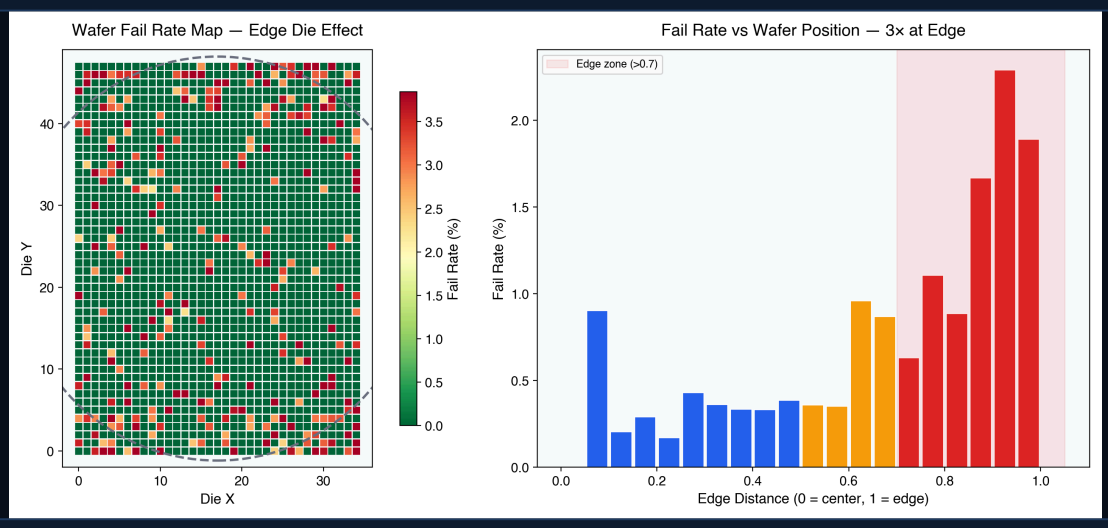
What you see: Left: throughput. A100 delivers 88,128 samples/s, 49x faster than CPU (2,000/s). Right: time for 50 epochs. A100 completes in 1.6 hours, CPU would take 69.2 hours. For automated production retraining on Day 30, waiting 3 days for CPU is not an option.

88K/s	49x	219 min
A100 throughput (bfloat16)	A100 vs CPU speedup	40-day pipeline total wall clock

Production path: **AWS g4dn.xlarge (T4, $0.526/hr)** for scheduled retrains. A100 Colab for initial training. Auto-stop script terminates EC2 after each run.

RESULTS

200M rows. 40 days.
The pipeline held.



Final result: 200M rows processed, 8.5 GB Parquet on S3, 219 minutes wall clock on A100. 1 automated retrain (Day 30). 24 critical drift days. 1 canary failure + automatic rollback (Day 39). System self-healed with zero human interventions.

The model is rarely the bottleneck in production ML.

It is the drift strategy, the retrain gate, and the rollback safety net that keep it reliable at day 40.

`github.com/rajendarmuddasani/DRAM_Yield_Predictor_MLOps`

#MLOPS #SEMICONDUCTOR #DRAM #PYTORCH